Relating Fractions, Decimals, and Percentages

Decimals, fractions, and percents are all ways to express numbers as amounts other than whole numbers. Decimals use digits to the right of the decimal point to write tenths, hundredths, thousandths, and so on. In fractions, the numerator expresses the number of parts and the denominator expresses the whole. A percent indicates the number of parts out of 100.

Example 1

Hide

Write 80% as a decimal.

Step 1

Drop the percent sign and move the decimal point two places *to the left*.

80% = .80

(*Note:* 80% = 80.%)

Step 2

Drop any unnecessary zeros.

.80 = **.8**

Example 2

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What is .3 expressed as a percent?

Step 1

Move the decimal point two places to the right, adding any necessary zeros.

.30.

Step 2

Add a percent sign.

.30 = **30%**

Example 3

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Emily finished 25% of her paperwork. What fraction of her paperwork did she finish?

Step 1

Drop the percent sign and write the number with a denominator of 100.

$25\% = \frac{25}{100}$

Step 2

Simplify the fraction if necessary.

$\frac{25}{100} \div \frac{25}{25} = \frac{1}{4}$

Emily finished $\frac{1}{4}$ of her paperwork.

Example 4

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$\frac{4}{5}$ of the registered voters of Barden voted in the mayoral election. What percent of the registered voters voted?

Step 1

Divide the denominator into the numerator.

$$5\overline{)4.0}$$

Using a calculator:

4 ÷ 5[enter] 4÷5 0.8

Step 2

Change the decimal answer into a percent by moving the decimal point two places *to the right*.

.80.

Step 3

Add a percent sign.

.80 = 80%

80% of the registered voters of Barden voted in the election.

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